A closed-form variance floor for slope-capped diffusion samplers on thin manifolds

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Claim. For data in a σ-tube about a manifold, any reverse-SDE sampler whose learned normal score slope is bounded by $\bar{\kappa}$ retains normal variance at least $1/(8\bar{\kappa}^2)$ as $\sigma \to 0$, for *every* truncation time and *every* schedule. So a slope-capped model cannot track a thinning manifold, and bounded relative error forces $\bar{\kappa} \gtrsim 1/\sigma$, i.e. a slope budget growing like $\sqrt{J}$ in the Fisher information. This bounds the network's normal Lipschitz norm, not its parameter count: a fixed architecture attains any $\bar{\kappa}$ by enlarging its weights. This note is self-contained; §3 is the entire argument (comparison principle, integrating factor, one Gamma integral).

1 Setup

Let $M \subset \mathbb{R}^d$ be a compact smooth k-manifold with reach $\tau > 0$ and $X_0 = \varphi(U) + \sigma N_\perp$, U uniform on M, $N_\perp$ standard normal in the normal space. Use the variance-preserving OU process $X_t = aX_0 + sZ$, $a = e^{-t/2}$, $s^2 = 1 - e^{-t}$. Work in one normal coordinate $x \in \mathbb{R}$, treating the fiber marginal as Gaussian and decoupled from tangential position (*flat-tube idealization*; curvature enters at $O(s^2/\tau)$). Then $X_0 \sim \mathcal{N}(0, \sigma^2)$, $X_t \sim \mathcal{N}(0, \gamma^2)$ with $\gamma^2 = a^2\sigma^2 + s^2$, and the denoising target is exactly linear:

$$\varepsilon^*(x,t) = \kappa(t)x, \qquad \kappa(t) = s(t)/\gamma^2(t). \tag{1}$$

Maximising (1) over $u := s^2$ gives $u^\star = \sigma^2/(1-\sigma^2)$ and

$$\kappa_{\max} = \tfrac{1}{2\sigma\sqrt{1-\sigma^2}} = \tfrac{1}{2\sigma}(1 + O(\sigma^2)). \tag{2}$$

This is the mechanism: κ rises from 0, peaks at $\approx 1/(2\sigma)$ near $t \approx \sigma^2$, and returns to 0 as $t \to 0$. The score is steep only inside a short, finite window of noise levels, and the peak diverges as the tube thins.

Let X_t be the sampler's *own* ensemble at noise level t and define the realised normal slope

$$\hat{\kappa}(t) := \operatorname{Cov}\big(X_t, \hat{\varepsilon}(X_t, t)\big)\big/\operatorname{Var}(X_t),$$

the L^2 regression slope of the estimator under the law the sampler actually visits. No decomposition of the network is needed. The only hypothesis used below is

$$\hat{\kappa}(t) \le \bar{\kappa} < \infty \quad \text{for all } t, \tag{3}$$

a finite normal Lipschitz constant. This bounds the network's normal *slope*, a weight-norm quantity, not its parameter count: a fixed architecture realises arbitrarily large $\bar{\kappa}$ by scaling its weights, so nothing below lower-bounds network size. Put $\rho := 2\sigma\bar{\kappa} = \bar{\kappa}/\kappa_{\max} + O(\sigma^2)$, the single dimensionless knob: how close the network's ceiling gets to the peak slope the data demands.

2 Two structural facts

Lemma 1 (variance ODE). *Under the reverse Euler scheme $x \leftarrow x + (-\frac{1}{2}x + \hat{\varepsilon}/s)dt + \sqrt{|dt|}\,\xi$ with affine estimator, the ensemble normal variance obeys, in the small-step limit,*

$$-\frac{dV}{dt} = \Big(1 - \frac{2\hat{\kappa}(t)}{s(t)}\Big)V + 1. \tag{4}$$

Check: with the exact slope $\hat{\kappa} = \kappa$, $V(t) = \gamma^2(t)$ solves (4) identically. The forcing $+1$ is the sampler's own injected noise.

Lemma 2 (Lipschitz sufficiency)**.** *If $\hat\varepsilon(\cdot,t)$ is $\bar\kappa$-Lipschitz in the normal coordinate then $\hat\kappa(t)\le\bar\kappa$ under any law: for an i.i.d. copy X' of X,* $\mathrm{Cov}(X,g(X))=\frac{1}{2}\mathbb{E}[(X-X')(g(X)-g(X'))]\le\frac{\bar\kappa}{2}\mathbb{E}[(X-X')^2]=\bar\kappa\,\mathrm{Var}(X)$*. Lipschitz control is sufficient, not necessary.*

This is why the floor is not an artifact of linearising the score. (4) is an *identity* for an arbitrary nonlinear $\hat\varepsilon$: for the continuous reverse SDE $dX=f(X)d\tau+dW$ with $f(x)=\frac{1}{2}x-\hat\varepsilon(x,t)/s$, Itô gives $\mathrm{Var}'=2\,\mathrm{Cov}(X,f(X))+1$, which is (4). The nonlinearity is not discarded, it is absorbed into the definition of $\hat\kappa$.

Singular rescaling. Put $t=\sigma^2u$, $V=\sigma^2W$, $\hat\kappa=(1/\sigma)m(u)$. Since $s=\sqrt{t}(1+O(t))$, (4) becomes, as $\sigma\to0$ at fixed ρ,

$$-\frac{dW}{du}=1-\frac{2m(u)}{\sqrt{u}}W, \tag{5}$$

which is σ-free. The uncapped equation $-dW/du=1-2W/(1+u)$ has general solution $W=(1+u)+A(1+u)^2$, so deviations from the slave $W=1+u$ contract like $((1+u)/(1+u'))^2$ under downward integration: the band-entry boundary layer is suppressed quadratically. Hence $\lim_{t_0\to0}V(t_0)/\sigma^2=g(\rho)(1+O(\sigma^2/\rho^2))$, a *relative* correction.

3 The unconditional floor

Theorem 1. *Assume only (3). Then for every $t_0>0$ and every schedule,*

$$V(t_0)\ \ge\ \sigma^2\,\frac{1}{2\rho^2}(1+o(1))\ =\ \frac{1}{8\bar\kappa^2}(1+o(1))\qquad(\sigma\to0), \tag{6}$$

and the floor exceeds the true width σ^2 exactly when $\rho<1/\sqrt{2}$.

Proof. (i) Comparison. In (4) forcing and data are common and $\partial(-dV/dt)/\partial\hat\kappa=-2V/s\le0$: smaller slope gives larger variance. So any profile with $\hat\kappa\le\bar\kappa$ satisfies $V\ge V_{\hat\kappa\equiv\bar\kappa}$ pointwise. This is one-sided; no lower bound on $\hat\kappa$ is used, so arbitrary overshoot is permitted. *(ii) Envelope.* Take $m\equiv\rho/2$ in (5); the integrating factor $e^{-2\rho\sqrt{u}}$ gives $\frac{d}{du}[We^{-2\rho\sqrt{u}}]=-e^{-2\rho\sqrt{u}}$. *(iii) Integrate.* For data of sub-quadratic growth at $u=\infty$,

$$W(0^+)=\int_0^\infty e^{-2\rho\sqrt{u}}du=2\int_0^\infty xe^{-2\rho x}dx=\frac{1}{2\rho^2},$$

exactly, for every $\rho>0$. Undoing the rescaling gives (6); $1/(2\rho^2)>1\iff\rho<1/\sqrt{2}$. It applies to the full nonlinear estimator with no extension step, since (4) is exact. ☐

Refinement (no overshoot). If also $\hat\kappa\le\kappa$ (the network never over-contracts), then $V(t_0)\ge\sigma^2g(\rho)(1+o(1))$ with, for $q=\sqrt{1-\rho^2}$, $x_\mp=(1\mp q)/\rho$, $u_\mp=x_\mp^2$,

$$W_{\text{exit}}=(1+u_+)e^{-4q}+\tfrac{(3-2q)-(3+2q)e^{-4q}}{2\rho^2},\qquad g(\rho)=1+\tfrac{W_{\text{exit}}-(1+u_-)}{(1+u_-)^2}.$$

Band edges solve $\sqrt{u}/(1+u)=\rho/2$, i.e. $\rho x^2-2x+\rho=0$ ($x=\sqrt{u}$), nonempty iff $\rho<1$; above the band $W=1+u$ exactly, across it the same integrating factor with antiderivative $-(2\rho\sqrt{u}+1)e^{-2\rho\sqrt{u}}/(2\rho^2)$ gives W_{exit}, and below it matching $(1+u)+A(1+u)^2$ at u_- gives $g=1+A$. Here $g>1$ on $(0,1)$ and $g=C_0/\rho^2+O(1)$ as $\rho\to0$ with $C_0=(1+3e^{-4})/2$, so the floor is $0.1319\,\bar\kappa^{-2}$. The two constants, $1/8$ and $(1+3e^{-4})/8$, differ by 5.5%.

4 Consequence, assumptions, and how to refute it

Hold $\bar\kappa$ fixed and let $\sigma\to0$, so $J\asymp(d-k)/\sigma^2\to\infty$. Then $V/\sigma^2\ge(1+o(1))/(8\bar\kappa^2\sigma^2)\to\infty$: a sampler held at a fixed slope ceiling collapses without bound relative to the data. Inverting, bounded relative error needs $\bar\kappa\gtrsim1/\sigma$, i.e. a slope budget $\sim\sqrt{J}$. Since the bound holds for every t_0 and every schedule, no annealing or truncation schedule evades it. Whether a given architecture supplies that slope budget is a separate question, not settled here.

Assumed (where a sceptic should push): (1) the flat-tube idealization, one normal coordinate with Gaussian fiber decoupled from tangential position, curvature entering at $O(s^2/\tau)$ and not carried in the constants; (2) $\hat\kappa$ is the regression slope under the sampler's own ensemble, so a probe that measures it under p_t instead estimates a different quantity whenever $\hat{\varepsilon}$ is nonlinear; (3) the $o(1)$ in (6) is the $\sigma \to 0$ limit, relative correction $O(\sigma^2/\rho^2)$, so the additive gap to g is $O(\sigma^2/\rho^4)$.

Checks. At https://github.com/Naman84789/model-collapse-manifolds, `falsify_floor.py` attacks the result *from* (4) (RK45 vs. both closed forms; limits $\rho \to 1^-$, $\rho \to 0$, $\bar{\kappa} \to \infty$; 200 random slope profiles including overshooting ones; boundary-layer contraction; symbolic re-derivation). `e2e_floor_check.py` attacks it *before* (4): it recovers (2) from the density, then simulates the reverse SDE directly with a slope-capped score and measures the output variance without using (4) at all, recovering $1/(2\rho^2)$ to 1.7% at the equality case.

To refute: exhibit a profile with $\hat{\kappa} \le \bar{\kappa}$ whose measured $V(t_0)$ falls below $\sigma^2/(2\rho^2)$.